Project Guidelines: Scientific Ignorance Discovery in [Selected Field]

Project Overview

This project challenges students to replicate and extend the **GAPMAP** framework—a system designed to identify both explicit (stated) and implicit (context-inferred) research gaps. For graduate-level readiness, students must move beyond simple summarization to **abductive reasoning**, using LLMs to infer what a dataset *fails* to explain about its domain.

Part 1: Literature Survey and Non-Biomedical Field Selection

Goal: Students identify a high-impact research field and a corresponding large-scale neuroimaging or behavioral dataset.

- **Task:** Survey the **GAPMAP** paper's definitions of explicit and implicit gaps. Identify a target field.
- **Deliverable:** A 3-page survey identifying the "Grounds" (existing evidence) in the chosen field and the "Implicit Gaps" that currently exist.

Part 2: Baseline Reproduction & Explicit Gap Extraction

Goal: Implement the "Explicit Gap" extraction pipeline using a local or API-based LLM.

- **Task:** Reproduce the **ROUGE-L F1** similarity benchmarking described in GAPMAP. Students must feed the LLM chunks of their chosen dataset (e.g., experimental descriptions from THINGS-data) to extract explicit uncertainty markers like "it remains unknown" or "further research is needed".
- **Metric:** Report precision, recall, and F1 scores against a manually annotated "gold standard" subset (at least 50 samples).
- **Deliverable:** A Python Notebook and a progress report showing the overlap between different models (e.g., Llama-3 vs. GPT-4o).

Part 3: Final

- Students prepare final report, presentation and poster.

Grading Rubrics

Rubric 1: Part 1 - Field Survey & Theory

Criteria	Excellent (90-100%)	Satisfactory (70-89%)	Needs Work (<70%)

Gap Categorization	Correctly distinguishes between explicit lexical cues and implicit context gaps.	Identifies gaps but struggles to classify them as implicit vs. explicit.	Only identifies general research topics, not specific gaps.
Dataset Choice	Selects a high-resolution dataset (e.g., THINGS-MEG/fMRI) with a clear application.	Dataset is appropriate but lacks the complexity for implicit inference.	Dataset is too small or lacks the necessary metadata for "Grounds".

Rubric 2: Part 2 - Technical Baseline

Criteria	Excellent (90-100%)	Satisfactory (70-89%)	Needs Work (<70%)
Benchmarking Rigor	Correctly uses ROUGE-L with stemming and a 0.55 threshold.	Implementation is correct but lacks the 1,000-word chunking strategy.	Major errors in calculating F1 or precision/recall.
Model Comparison	Compares at least one open-weight (Llama/Gemma) and one closed-weight model.	Only uses one model type.	No quantitative comparison provided.

Rubric 3: Part 3 - Innovation & Results

Criteria	Excellent (90-100%)	Satisfactory (70-89%)	Needs Work (<70%)
Interpretability	Visualizes regional importance weights.	Shows basic loss curves but no biological analysis.	No analysis of what the model learned.

Scientific Output	Paper follows NeurIPS/ CVPR style with a clear "Limitations" section.	Professional report but lacks a technical poster.	Unstructured report.